LAB 16 – Elastic Net Regression

2403A52006

MD.Mustafa

Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, ElasticNet
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

Load Dataset

```
df = pd.read_csv("/content/insurance.csv")

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

Encode Categorical Variables

```
df_encoded = pd.get_dummies(df, drop_first=True)

df_encoded.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest
0	19	27.900	0	16884.92400	False	True	False
1	18	33.770	1	1725.55230	True	False	False
2	28	33.000	3	4449.46200	True	False	False
3	33	22.705	0	21984.47061	True	False	True
4	32	28.880	0	3866.85520	True	False	True

Next steps: Generate code with df_encoded New interactive sheet

Select Features and Target

```
X = df_encoded[['age', 'bmi', 'children', 'smoker_yes']]
y = df_encoded['charges']
```

Train/Test Split (80/20)

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Feature Scaling

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Linear Regression

```
lin_model = LinearRegression()
lin_model.fit(X_train_scaled, y_train)

y_pred_lin = lin_model.predict(X_test_scaled)

mae_lin = mean_absolute_error(y_test, y_pred_lin)
mse_lin = mean_squared_error(y_test, y_pred_lin)
r2_lin = r2_score(y_test, y_pred_lin)

print("Linear Regression Results")
print("MAE:", round(mae_lin,2))
```

```
print("MSE:", round(mse_lin,2))
print("R2:", round(r2_lin,4))
```

```
Linear Regression Results
MAE: 4213.8
MSE: 33981653.95
R2: 0.7811
```

Elastic Net with Different Alpha & l1_ratio

```
results = []

alphas = [0.1, 1, 10]
l1_ratios = [0.2, 0.5, 0.8]

for alpha in alphas:
    for l1 in l1_ratios:

        enet = ElasticNet(alpha=alpha, l1_ratio=l1, random_state=42)
        enet.fit(X_train_scaled, y_train)

        y_pred = enet.predict(X_test_scaled)

        mae = mean_absolute_error(y_test, y_pred)
        mse = mean_squared_error(y_test, y_pred)
        r2 = r2_score(y_test, y_pred)

        results.append([alpha, l1, mae, mse, r2])

results_df = pd.DataFrame(
    results,
    columns=["Alpha", "L1 Ratio", "MAE", "MSE", "R2"]
)

results_df
```

	Alpha	L1 Ratio	MAE	MSE	R2
0	0.1	0.2	4358.630440	3.509125e+07	0.773968
1	0.1	0.5	4299.391457	3.454483e+07	0.777487
2	0.1	0.8	4246.829731	3.414796e+07	0.780044
3	1.0	0.2	5801.679643	5.991952e+07	0.614042
4	1.0	0.5	5277.376653	4.914671e+07	0.683432
5	1.0	0.8	4625.086678	3.831187e+07	0.753223
6	10.0	0.2	8765.797806	1.308241e+08	0.157326
7	10.0	0.5	8355.405945	1.195752e+08	0.229783
8	10.0	0.8	7154.287032	8.991735e+07	0.420817

Next steps: Generate code with `results_df` New interactive sheet

Find Best Model

```
best_model = results_df.loc[results_df['MSE'].idxmin()]

print("Best Elastic Net Model:")
print(best_model)
```

```
Best Elastic Net Model:
Alpha       1.000000e-01
L1 Ratio    8.000000e-01
MAE         4.246830e+03
MSE         3.414796e+07
R2          7.800436e-01
Name: 2, dtype: float64
```

Compare Linear vs ElasticNet

```
print("\nModel Comparison")

print("\nLinear Regression:")
print("MAE:", round(mae_lin,2))
print("MSE:", round(mse_lin,2))
print("R2:", round(r2_lin,4))

print("\nBest ElasticNet:")
print(best_model)
```

```

Model Comparison

Linear Regression:
MAE: 4213.8
MSE: 33981653.95
R2: 0.7811

Best ElasticNet:
Alpha       1.000000e-01
L1 Ratio    8.000000e-01
MAE         4.246830e+03
MSE         3.414796e+07
R2          7.800436e-01
Name: 2, dtype: float64
```

Conclusion

Elastic Net combines both L1 (Lasso) and L2 (Ridge) regularization. It helps in handling multicollinearity and improves model generalization. The best model is selected based on lowest MSE and highest R^2 score. Compared to Linear Regression, Elastic Net may reduce overfitting